# PARAKH — Stage 7 PRD

## Explainability, Interface & Demo Layer

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## 1. Objective

Transform system outputs into a **clear, interactive, human-understandable interface** that:

- Explains *why* a record was flagged
- Shows **confidence vs risk separation visually**
- Enables **quick inspection and storytelling**

This stage is critical for:

- SIH judging
- usability
- trust

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## 2. Scope

### Included

- CLI or simple UI (web or notebook)
- Record-level explanations
- Summary dashboards
- Filtering & inspection tools
- Visualization of scores

### Excluded

- Core computation (Stages 1–6)
- Model changes

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## 3. Inputs

From previous stages:

```
Corpus
ConfidenceResult
RiskFeatureResult
RiskResult
DecisionResult
```

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## 4. Outputs

User-facing system that allows:

- Viewing records
- Filtering by decision
- Inspecting explanations
- Visualizing distributions

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## 5. Interface Options

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### Option A (Recommended for SIH)

**Streamlit Web App**

Fast, simple, demo-friendly

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### Option B

**Jupyter Notebook Dashboard**

Simpler but less polished

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### Option C

**CLI Tool**

Backup/demo fallback

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## 6. Core Features

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### 6.1 Record Table View

Display:

| work_id | work_name | district | C | R | decision |
|---|---|---|---|---|---|

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### 6.2 Filters

User must be able to filter by:

- decision type (INVESTIGATE / REMEDIATE / etc.)
- confidence range
- risk range
- district

## 6.3 Record Drill-Down (MOST IMPORTANT)

Clicking a record shows:

```
Work: Road construction in X district

Decision: INVESTIGATE

Confidence: 0.82
Risk: 0.91

Why flagged:
- Cost is 3.2x higher than peer median
- Vendor concentration is high (HHI = 0.76)
- Similar work found within 30 days (duplicate score = 0.88)

Data quality:
- All fields present
- Dates valid
- Amounts consistent
```

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## 6.4 Confidence vs Risk Plot

Scatter plot:

- X-axis → Confidence
- Y-axis → Risk

Quadrants:

- Top-right → INVESTIGATE
- Bottom-left → CLEAR
- Bottom-right → MONITOR
- Top-left → (rare)

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## 6.5 Distribution Charts

- Histogram of confidence
- Histogram of risk
- Decision breakdown (bar chart)

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## 6.6 Summary Metrics

```
{
  "total_records": N,
  "investigate_count": X,
  "remediate_count": Y,
  "avg_confidence": ...,
  "avg_risk": ...
}
```

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# 7. Explainability Engine

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## 7.1 Top Signal Extraction

For each record:

- Identify top contributing signals:
    - highest z_cost
    - high HHI
    - high burst
    - high duplicate score

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## 7.2 Template

```
reason = f"""
High risk due to:
- Cost anomaly (z = {z_cost:.2f})
- Vendor concentration (HHI = {hhi:.2f})
- Temporal burst detected

Confidence is {C:.2f}, indicating reliable data.
"""
```

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## 7.3 Rules

- Max 4 bullet points
- Plain English only
- No math symbols in UI

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# 8. Implementation Architecture

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## 8.1 Modules

- `ui_app.py` (Streamlit)
- `visualization.py`
- `explain.py`

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### 8.2 Example (Streamlit)

```
import streamlit as st

st.title("PARAKH Dashboard")

df = load_results()

decision_filter = st.selectbox("Decision", df["decision"].unique())
filtered = df[df["decision"] == decision_filter]

st.dataframe(filtered)

selected = st.selectbox("Select Record", filtered["work_id"])

record = df[df["work_id"] == selected].iloc[0]

st.write(record["explanation"])
```

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## 9. Non-Functional Requirements

### Speed

- UI loads < 2 seconds

### Clarity

- Non-technical user can understand output

### Stability

- No crashes during demo

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## 10. Demo Flow (CRITICAL)

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### Step 1

Show dataset overview

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### Step 2

Show confidence distribution

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### Step 3

Show risk vs confidence plot

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### Step 4

Click INVESTIGATE record

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### Step 5

Explain:

- why it's risky
- why data is trustworthy

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### Step 6

Show REMEDIATE case:

- "we don't trust this data"

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## 11. Acceptance Criteria

Stage complete if:

- [ ] UI displays all records
- [ ] Filters work correctly
- [ ] Record explanations readable
- [ ] Charts render correctly
- [ ] Demo flow works smoothly

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## 12. Definition of Done

- Fully working demo interface
- End-to-end system visible
- Ready for SIH presentation

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## 13. Key Design Principle

> **If you can't explain it, it doesn't exist.**

The system is not judged by:

- math
- models

It is judged by:

- clarity
- trust
- explanation

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